

Asynchronous transfer mode — an overview

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A review is made of some of the early study items of ATM, as a way of introducing its terminology and concepts. Building upon these concepts are new study activities developing specialised ATM-based services together with supporting network functions. An overview of some of these activities is given and several of these studies are the subject of the others papers in this issue of the BT Technology Journal. The theme of this paper concentrates on the fact that ATM is beginning to realise its promise of being a high performance, ubiquitous, flexible transport mode and a stimulus for new innovation.

1. Why asynchronous transfer mode (ATM)?

The era is upon us of broadband networks and specialised services supported by ATM equipment. Public ATM networks currently offer permanent virtual circuits (PVCs) and permanent virtual paths (PVPs). These can support circuit emulation with a large variety of rates, or provide a flexible backbone for a virtual private network where the backbone is shared by traffic from a variety of applications. It is expected that the roll-out of switched services (switched virtual circuits — SVCs) on public networks sometime soon after inter-exchange signalling standards are agreed around the beginning of 1996. (Recommendation Q.2722.1, B-ISUP including point-to-multipoint, is expected to be frozen and submitted for accelerated approval in October 1995. Recommendation Q. 2725.1/2, B-ISUP including more advanced capabilities for negotiation of parameters, is expected to be submitted for accelerated approval in February 1996.)

Specialised ATM services are already being developed. An example is LAN emulation where, through a server function, the sending end-system determines the ATM-layer address of the receiving end-system, given only its 802.3/5 MAC-layer address. It then uses or establishes a direct VC to forward data. An early implementation may route multiple VCs between the end-systems of different sites via a PVP across the public network.

As another example of what is currently being developed, there is a new ATM-layer bearer service, known as the available bit rate (ABR) service, which provides fast ‘intelligent’ management of bandwidth on a data VP or VC where the traffic activity level is continually varying. This is likely to provide an early focus for enhanced service offerings associated with permanent and switched circuits.

Some of these terms, which are used throughout this issue of the BT Technology Journal, may be unfamiliar to

the reader. Therefore it is the purpose of this introductory paper to set the scene and describe some of the fundamental concepts associated with ATM.

First of all, how did the dream of ATM turn into the success story described above? In other words, why ATM? It is a worthwhile exercise to look at this question, at least briefly, because it helps to set the context for the research papers in this issue of the Journal.

The concepts of ATM and its predecessor, asynchronous time division (ATD), have been around since the early 1980s. Some very early pioneering work was performed at CNET’s laboratories in Lannion, France on project Prelude. It was based on the notion that voice, video and data streams could all be transported and switched as fixed-length cells. These cells were the precursor of the current ATM cell (see Fig 1).

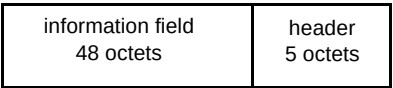


Fig 1 The ATM cell.

Even in those early days, the essential properties of ATM were clearly understood:

- it could support any mixture of real time and non-real time connections while guaranteeing their different quality of service requirements,
- a huge range of service bit rates could be supported by the ATM switches.

In a word it provided flexibility — as the basis of a network which can be used to support all kinds of new services and which anticipates that increasing speed is

going to be demanded as the realms of the information superhighway approach.

Of course it is possible today to achieve huge transmission rates over a single optical fibre. But ATM provides the key to unlocking the contents of that bit stream and processing each individual connection in the total flow. Thus, the total flow of cells arriving at a switch port divides into hundreds or thousands of separate connections, each one separately routed through the switch and beyond. Each connection may include characteristics like bursty flows or steady flows at different rates. ATM provides the switching and traffic control functions necessary to support this complex problem.

But technical wizardry is not a complete answer to why the world's network providers, switch manufacturers and end-system manufacturers are moving to ATM. A significant part of the success story is due to the formation of interest groups which served to promote and shape ATM for the market-place. In particular, the work of the ATM Forum has been fundamental to the acceptance of the technology and is a pointer to how future network technology can be successfully introduced. This topic is discussed more fully later in the paper.

2. Early problems

Some people nowadays dismiss the work done on ATM during the 1980s as the slow deliberations of a small technical elite, with no proper commercial direction. But there were many serious problems to address and the ATM bandwagon seen today would never have come into being if the technical foundations were not solid. The ITU-T (then known as the CCITT) and the RACE programme served us well to promote the careful investigations which have led to sound technology.

2.1 The cell header

It took a long time to get the fundamentals agreed like the size of the ATM cell and the structure of the header part of each cell. The key element of the ATM header is the virtual connection identifier (VCI). It is supplemented by a second field, the virtual path identifier (VPI), so that the values in this pair of fields identify a single connection within the total flow of cells arriving at a switch port. Most

of the properties of each connection can be found through look-up tables in each switch (e.g. real time, or non-real time). Only very limited additional information is carried in the cell header. The additional information is mainly to allow different types of cell or control settings to be distinguished within the flow of a single connection (e.g. high/low priority cells, OAM cells, explicit forward congestion indication, resource management control cells). For more on the use of some of these header functions, see Adams and Smith [1].

The concept of a virtual path was not present in the original ATD proposals. There was a lengthy debate on the need for additional bits in the cell header to allow for a VPI. The more bits in the header, the lower the efficiency of bandwidth utilisation. Also some argued that the huge VPI/VCI address space (24 bits on each interface linking any two public switches) would never be justified and would simply add unwarranted cost to these interfaces — but it has a pay-off in terms of reducing network management complexity. In essence it is a container for hundreds or thousands of separate VCs which are all to be routed the same way (i.e. following the route of the virtual path — see Fig 2).

In Fig 2 there are network elements called VP cross-connects that only process the VP field to determine the onward routing. They are unaware of different VCs within a VP and, hence, VCs can be set up and cleared down without ever informing the VP crossconnects. This reduces the network management problem for establishing a large number of PVCs between two distant sites.

A crossconnect at the physical layer could also ease network management problems in the same way. But the benefit of a VP is that it offers a very large range of possible rates and at least two classes, e.g. low priority or 'best effort', and high priority appropriate for circuit emulation. Also the VP bandwidth can expand or shrink in line with customer activity and this becomes particularly effective if ABR controls are employed [1].

2.2 Traffic controls

There has to be some notion of the bandwidth to be associated with a VC at each VC switch, and similarly there has to be a notion of the bandwidth to be associated with a

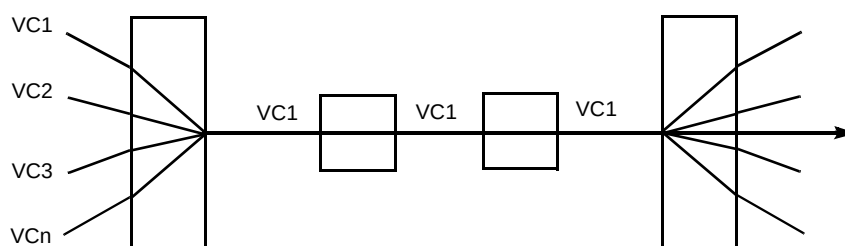


Fig 2 An ATM virtual path.

VP at the VC switches and the VP crossconnects. The term 'bandwidth' here does not imply some fixed partitioning of physical links. It has to make allowance for the fact that several ATM cells belonging to one VC or VP can cluster together and very temporarily occupy most or all of the physical link capacity. This clustering effect arises naturally due to delay variations within the ATM switches and multiplexers.

Traffic control methods have been an item of study since the earliest days. There were difficult issues to overcome and some issues, like the impact of cell-delay variations, were not very well understood at the beginning. The degree of difficulty and the number of solutions which had to be evaluated is the main reason why the first substantial ITU-T Recommendation on ATM traffic matters (I.371) [2] was not completed until 1992.

Recommendation I.371 is currently being revised because the first issue only dealt with controls related to the peak rate. The peak rate, as its name suggests, is the maximum rate expected on a connection (after averaging out the clustering effect) and this provides no direct information or control parameters to assist statistical multiplexing. Even so, this was a major achievement and is the basis for the traffic control of today's ATM networks. For example, it provided the key framework for the way in which ATM connections will be policed by the network.

This introduces yet another term and another one of the early problems — policing. Policing is a necessary function at the edge of a public network to ensure that each connection keeps within agreed traffic load limits. Without that, any excess load may cause quality of service degradation. A relatively simple algorithm had to be devised which allowed policing to be performed simultaneously on many thousands of separate connections at each input port of a local ATM exchange. It had to permit each connection to have some very limited level of cell clustering (because this is unavoidable), while checking that the average rate does not exceed the bandwidth resources available to it. The policer can delete cells which exceed the rate limit.

The achievement of an internationally agreed standard reference algorithm for the policer was a major milestone. The complex patterns of a connection's cell arrivals at a local exchange were reduced to standard parameters which were simple to monitor. Each time a cell arrived the monitor would increment a counter. The counter would always be decremented at the true rate at which cells should arrive. No cells would be discarded by the policer provided the counter remained at or below some small value representing the permissible level of clustering.

These concepts led to further discussions on the possible need for shaper functions in large private networks, or

between two public networks. The job of the shaper would be to remove excessive clustering and thus ensure that cell losses are avoided at the policer. The need for shapers depends on the tolerance of the policer which, in turn, has to be tuned according to the capabilities of the switch buffers to absorb large cell clusters. There has been much study of this subject, together with the general problem of determining the bandwidth and buffer requirements of call requests, given limited information about the source traffic behaviour (see Key [3]).

2.3 *The ATM adaptation layer (AAL)*

No discussion of early problems would be complete without mentioning the ATM adaptation layer.

Given the vision (set at the time when ATM was called ATD) of a cell-based network supporting a mix of real time and non-real time services, something more was needed than the basic functions supported by ATM cells with their simple headers. Additional functions must be provided which are special to data services, e.g. the location of packet boundaries within a set of cells. More functions must be provided which are special to real time services, e.g. timing information which allows the sender to directly control the constant rate used by the distant end to forward information octets towards the application.

Because this is specialised, service-related information, it would be very inefficient to create additional cell-header octets that catered for all possible needs. Therefore another layer was devised which provided for both common functions (like segmentation of the data stream into cell-sized blocks, and re-assembly), and service-specific functions supported by an AAL header and sometimes a trailer.

Referring to Fig 3, a small number of AALs were defined, sufficient for the needs of a very large range of services.

AAL3/4 was originally two AALs — AAL3 was aimed at connection-oriented data and was a candidate for signalling, while AAL4 was aimed at connectionless (CL) data. However, there was a strong argument to reduce the number of different AALs which have to be supported in the end-systems and this led to them being combined. It was a popular opinion at one point that AAL3/4 would be the universal AAL for data transfer and signalling and that other AALs would not be required for the early marketplace.

This perhaps illustrates where the good work of the ITU-T needs to be supplemented by interest groups such as the ATM Forum. Moving up a layer from ATM, decisions become more involved with where a service-specific function may or may not be required, and this enters the

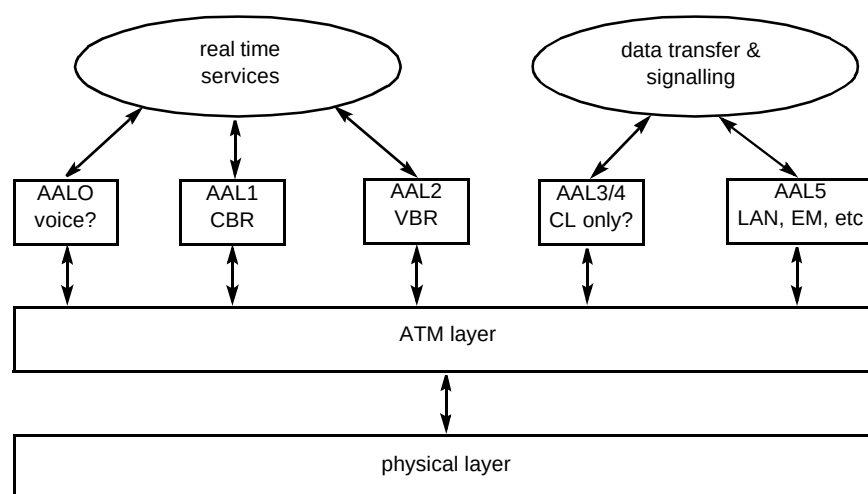


Fig 3 The different ATM adaptation layer categories.

world of the commercial realities of end-system manufacturers. AAL3/4 is very feature-rich (not least as a result of two AALs being combined into one) and this is a typical result of designing-in flexibility, not knowing which products are best suited to the demands of the market-place. The ATM Forum has done a magnificent job of getting the strong views of the end-system manufacturers to bear on what functions are necessary, what would add unnecessary cost or detract from a customer's perception of ATM, and so on. The result was a move away from AAL3/4, which requires a minimum of 4 octets in every cell, towards a proposed new, leaner AAL, called AAL5.

Incidentally, the argument was not just about AAL3/4. The same concerns were raised about AAL1 which is the main AAL for circuit emulation, suitable for video MPEG signals, etc. To get the ideal of a single AAL in the end-system, many people started looking at AAL5 for video. Video can be carried using the simplified AAL5, and some demonstrations showed this, but it has led to concerns about its suitability under conditions of varying network load (see Mulvey, Kim and Reid [4]).

AAL0 is simply a null AAL which some are now proposing for voice. All 48 octets of every cell would be available for voice samples. The inclusion of voice may, at first sight, seem a little odd in a discussion of broadband networks; however, voice will be one component of the traffic between two business sites which have established a PVP for flexible-rate circuit emulation. In this case the voice samples may be carried in the so-called composite format, i.e. each octet in the cell represents a different voice connection. To make better use of compression techniques (e.g. simple silence suppression) and allow greater use of the PVP bandwidth, it is better to use direct packetisation. In this case the information field is filled with the voice samples from one connection. Some form of echo control is needed because of the packetisation delay.

The ITU recommend AAL1 for voice telephony. It has the benefit of being fully developed and implemented, and therefore many question the proposal for AAL0 and see the quality benefits of AAL1 as important. For example, the sequence number field of AAL1 allows lost or misinserted cells to be detected which can be used to improve the quality at the receiver (for more on voice, see Alley, Kim and Atkinson [5]).

Finally, AAL2 is for variable bit rate video or voice. It is not yet finalised and it may differ very little from AAL1, so the need for this AAL is not yet clear.

3. Current problems

It would be nice to have a figure of how many man-hours of study and development have been invested on ATM world-wide so far. It has to represent a very substantial investment of time and resources and this only adds confidence to the expectation that ATM will be a very significant element of private and public networks in the near future. In the final analysis, any new network technology needs this 'snowball effect' of investment to make progress. In some ways the answer to 'Why ATM?' is simply that a new technology was needed for multiservices and multi-bit rates and ATM happened to be in the right place at the right time. This does not imply that other technologies are inferior; for example, the use of SMDS technology provides support for an excellent LAN-to-LAN service today. The SMDS service will continue to flourish because it is tuned to a particular market need. However, looking to the longer term, it is expected that further innovation and novel service concepts will be directed towards ATM, and this, more than anything, will determine the future volume share of ATM equipment. At some point it will make sense to migrate the SMDS service, and other services, to ATM (see also King et al [6]).

If all of this investment has occurred, why are the studies continuing at such a pace today? One answer must be that innovation is still taking place, primarily at the service level, together with associated studies on network functions, needed to support such services, e.g. ABR [1]. As suggested earlier, innovation is the long-term good-health factor for ATM.

At the physical layer there has been innovation in transmitting very high bit rates over unscreened twisted pair (UTP) for distances of 100 m or even greater. This has led to new proposals for the transmission of ATM over installed building wiring to achieve ATM to the desktop (see Thorne [7], which also looks at how ATM may be provided to the home).

Another innovation is the LAN emulation service. One version of this is shown in Fig 4.

The end-systems (workstations) have an ATM adapter card with an additional LAN emulation (LE) layer which takes 802.3 (Ethernet-type) or 802.5 (Token Ring-type) frames and forwards them towards the ATM hub. LAN emulation only deals with the cases of all-Ethernet or all-Token Ring virtual LANs (not mixed frame formats nor other formats like FDDI).

The frames are segmented into cell-sized blocks using AAL5. The route of most frames is along a direct VC set up by a signalling request from the source end-system supplying the ATM address of the destination end-system. This address (which may be a normal E.164 'telephone number-type' address, or an ATM Forum address format) is obtained by sending an address resolution request to the LE server. In Fig 4 the server is shown attached to the private ATM hub switch, possibly for service security reasons, but it may be managed by the network operator. A control PVC exists between any end-system and the LE server and this same PVC can be used to forward initial frames via the server while waiting for a direct VC to be established.

The use of multiple, separately routed VCs means that traffic bottlenecks caused by sending everything to the server is avoided. An initial service across a wide area could be provided where site-to-site VCs are established which have a common VP-part and follow a fixed route as shown in Fig 4. This is needed only while ATM public networks support just PVPs and PVCs. The ATM hubs could use a signalling VC channel over the same VP so that information is forwarded to the distant end to complete the connection establishment. For more information on the LAN emulation service, see Alley, Kim and Atkinson [5], and Barnes et al [8].

One new thing that is required of the network is that a connection request is rarely, if ever, refused. The server function has only a limited capacity as a data forwarder and it is preferred that a direct VC be established, even if this has only very little bandwidth. The best approach is that, as new VCs are established which travel, for example, through the VP shown in Fig 4, bandwidth is re-allocated so that all VCs get a reasonably fair share of the available capacity. This is what ABR provides (see Adams and Smith [1]).

4. Conclusions

Some of the ATM-related study issues, both of the past and of the present, have been described in this paper, which has acted as a vehicle for introducing some of the terminology and concepts that the reader may use as a background to the other papers in this issue of the BT Technology Journal. It has been, of necessity, a very brief introduction to the subject, and the reader new to ATM, and looking for more detail and some of the history, is advised to look, for example, at de Prycker [9].

There have been two phases in the development of ATM. Firstly, the work associated with the initial vision of a flexible future-proof network able to support current and future services both real time and non-real time (i.e. delay sensitive/insensitive), and, secondly, the work of the service

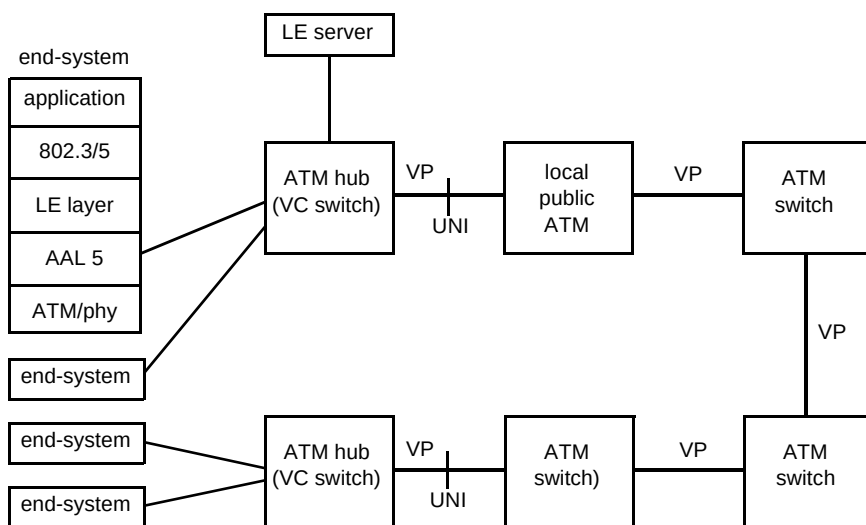


Fig 4 LAN emulation service utilising a PVP.

and application innovators of today, who are building on these sound foundations.

The current explosion of study issues surrounding ATM has come about because of the work of the ATM Forum promoting the involvement of end-system manufacturers and users. This good work includes their 'Ambassador' education programme on ATM. It has led to a whole new set of innovators who are relating ATM strongly to the market-place.

Looking to the future, it is not expected that the rate of arrival of new ideas will slow down. The last ten years or so have been an exciting time for anyone closely involved with the development of ATM. It has been a time when each of the many international meetings has led to a buzz of discussions on new thoughts and directions, and the next few years will see the appearance of many more new and exciting ideas in the ATM networks and services area.

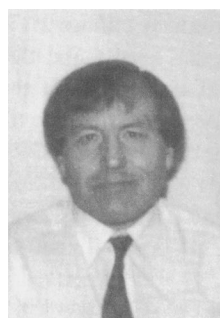
Appendix

List of acronyms

AAL	ATM adaptation layer
ABR	available bit rate (bearer service)
ATD	asynchronous time division
ATM	asynchronous transfer mode
B-ISUP	broadband-ISDN user part (inter-public exchange signalling)
CBR	constant bit rate
CCITT	International Telegraph and Telephone Consultative Committee
CL	connectionless
FDDI	fibre distributed data interface
ISDN	integrated services digital network
ITU-T	International Telecommunication Union — Telecommunication Sector
LAN	local area network
LE	LAN emulation
MAC	media access control
MPEG	Moving Picture Experts Group
OAM	operations and maintenance
PVC	permanent virtual circuit
PVP	permanent virtual path
RACE	Research into Advanced Communications in Europe
SMDS	switched multimegabit data service
SVC	switched virtual circuit
VBR	variable bit rate
VCI	virtual connection identifier
VPI	virtual path identifier
UTP	unscreened twisted pair
UNI	user/network interface

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John Adams received his PhD in Applied Mathematics from Leicester University in 1979.

He joined BT in 1971 and worked on overall grade of service evaluation for a while before joining the System X programme. There he was involved with testing and integration of local exchange equipment.

After leaving the System X programme, he went on to invent the Orwell Ring LAN protocol for carrying integrated voice, video and data on a shared slotted ring network. He then joined the broadband multiservice networks unit working primarily on ATM traffic control issues. He became an Engineering Advisor in that unit in 1990.

Currently, he heads an ATM standards committee, and leads several research projects within the unit investigating ATM networks and services.

